ClipSAT : Linear Algebra Vectors, Matrices, Eigenvalues & Least Squares [ANSWER KEY]

Unit 7.01: Linear Transformations & Change of Basis

Q1. Let Tmathbf x=Amathbf x with A=left[begin{matrix}-2 & -2-1 & 0end{matrix}right] . Find Tlangle 3, -3 rangle .

[x] A: langle 0, -3 rangle

[ ] B: langle -3, -6 rangle

[ ] C: langle 0, 3 rangle

[ ] D: langle 1, -2 rangle

Solution: Multiply: left[begin{matrix}-2 & -2-1 & 0end{matrix}right]left[begin{matrix}3-3end{matrix}right]=left[begin{matrix}0-3end{matrix}right] .

Q2. Which matrix represents the reflection across the line y = x?

[ ] A: left[begin{matrix}0 & -1-1 & 0end{ma

[x] B: left[begin{matrix}0 & 11 & 0end{matr

[ ] C: left[begin{matrix}0 & -11 & 0end{mat

[ ] D: left[begin{matrix}1 & 00 & 1end{matr

Solution: Find the images of mathbf e_1,mathbf e_2 : langle 0, 1 rangle and langle 1, 0 rangle . These are the columns.

Q3. Which map is linear?

[ ] A: Tx,y=x+1, y

[ ] B: Tx,y=xy, x

[ ] C: Tx,y=sin x, y

[x] D: TA=A^T text{on } M_{2times2}

Solution: TA=A^T text{on } M_{2times2} preserves sums and scalar multiples. The others fail: Tx,y=x+1, y T0 â‰ 0; Tx,y=xy, x xy is not linear; Tx,y=sin x, y sin is not linear.

Q4. Let Tmathbf x=Amathbf x with A=left[begin{matrix}-3 & -12 & -2end{matrix}right] . Find Tlangle -3, 2 rangle .

[ ] A: langle 13, -1 rangle

[ ] B: langle -7, 10 rangle

[ ] C: langle 8, -9 rangle

[x] D: langle 7, -10 rangle

Solution: Multiply: left[begin{matrix}-3 & -12 & -2end{matrix}right]left[begin{matrix}-32end{matrix}right]=left[begin{matrix}7-10end{matrix}right] .

Q5. A 3 times 3 symmetric matrix T_{a} has eigenvalues lambda_1 = 20 , lambda_2 = 22 , and lambda_3 = 20 . Find the exact value of the trace text{tr}T_{a} .

[x] A: 62

[ ] B: 40

[ ] C: 82

[ ] D: 65

Solution: The trace equals lambda_1 + lambda_2 + lambda_3 = 62 .

Q6. A linear map f: mathbb{R}^{20} to mathbb{R}^{20} has rank text{rank}f = 18 . Determine the exact value of the nullity dimker f .

[ ] A: 20

[ ] B: 5

[ ] C: 0

[x] D: 2

Solution: By Rank-Nullity, dimker f = 20 - 18 = 2 .

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Q7. Which matrix represents the orthogonal projection onto the x-axis?

[x] A: left[begin{matrix}1 & 00 & 0end{matr
[ ] B: left[begin{matrix}-1 & 00 & 0end{mat
[ ] C: left[begin{matrix}0 & 00 & 1end{matr
[ ] D: left[begin{matrix}0 & -11 & 0end{mat

Solution: Find the images of mathbf e_1,mathbf e_2 : langle 1, 0 rangle and langle 0, 0 rangle . These are the columns.

Q8. Which map is NOT linear?

[ ] A: Tx,y,z=x+z, 0
[x] B: TA=det A text{on } M_{2times2}
[ ] C: Tp=p1 text{on } P_3
[ ] D: TA=A^T text{on } M_{2times2}

Solution: TA=det A text{on } M_{2times2} is not linear: det2A = 4det A. The other three preserve sums and scalar multiples.